

A geographical unit that collects, stores and release water is called a watershed or catchment area. It's a land area that channels rainfall and snowmelt to creeks, streams, and rivers, and eventually to outflow points such as reservoirs, bays, and the ocean.

Landforms by River

*In rivers, erosion, transportation and deposition go on simultaneously from the point of origin to the point where they meet the sea. This process leads to the formation of different terrestrial relief features. Some of them are as follows-

Gorge - It is a deep narrow section of river valley usually with near vertical rock walls, more generally a narrow valley between hills and mountains. Gorges are formed as a result of continued erosion of mountainous terrain by a river. For Ex: **Chisapani Gorge** in Nepal is formed by Karnali River. **Karnali** is the longest river in Nepal (507 km). It is due to the high velocity of the river that it creates many gorges.

Canyon- Canyons are the more eroded version of gorges. **Grand Canyon** is located in Arizona (US). It is a steep-sided canyon carved by **Colorado River**. The canyon is 446/447 km long with an average width of 6.4 to 29 km and depth of 1.8 km deep.

V-shaped Valley- V-shaped valleys are formed by erosion. The majority of the rivers in the world carry on two processes simultaneously i.e. vertical and horizontal erosion respectively. These rivers create V-shaped valleys.

Waterfalls- A waterfall is a stream that falls from a height. Waterfalls are often the sites of the greatest concentration of energy dissipation along the course of a stream.

Meanders- As water flowing under gravity seldom flows straight for any long distance, a winding course soon appears. The irregularities of the ground force the river to swing in loops, forming meanders.

Alluvial Cones - Also called alluvial fans, they are cone-shaped deposits of sediments built by a stream where it merges into a broad valley or plain.

Delta- When the river reaches the sea, the fine materials it has not yet dropped are deposited at its mouth, forming a fan-shaped alluvial area called a delta. This alluvial tract is, in fact, a seaward extension of the floodplains. Deltas differ much in size, shape, growth and importance. A number of factors such as the rate of sedimentation the depth of the river, the sea bed and the character of tides, currents and waves greatly influence the eventual formation of deltas. The Ganges Delta is the largest delta in the world. It is also known as Ganges-Brahmaputra, Meghna Delta or Sunderban Delta.

1. Formation of delta is affected by :

- (a) Rocks
- (b) Tide - ebb
- (c) Strong winds
- (d) Deep sea

M.P.P.C.S. (Pre) 1993

Ans. (d)

Conditions that are favorable for the formation of deltas are as follows :

The river must carry a large amount of sediment.

The slope and speed of the river must not be strong enough to remove the sediments deposited by the river.

The slope of the sea floor or lake at the river mouth must be gentle or flat.

The large size and volume of the river.

Presence of a shallow adjoining sea or continental shelf.

Thus, a **river delta** is a landform that forms at the mouth of a river, where the river flows into an ocean, sea, lake, or reservoir. Deltas form with the deposition of sediments carried by a river. Over long periods, this deposition builds the characteristic geographic pattern of a river delta. Thus option (d) is the most probable answer.

2. The greatest delta of the world is formed by –

- (a) Ganga and Brahmaputra
- (b) Mississippi- Missauri
- (c) Yangtze-Kyang
- (d) Huang-Hoe

41st B.P.S.C. (Pre) 1996

Ans. (a)

The greatest delta in the world is the Ganges Delta. The Ganges Delta is a river delta in Bangladesh and West Bengal, India. It is also known as the Ganges–Brahmaputra-meghna Delta or Sunderban Delta. The Ganges Delta has the shape of a triangle and is considered to be an “arcuate” delta (arc-shaped). It covers approx. 105641 sq. km.

3. At the mouth of which of the following rivers the 'bird's foot' type delta is formed?

- (a) Huang Ho
- (b) Nile
- (c) Danube
- (d) Mississippi

U.P. P.C.S. (Pre) 2018

Ans. (d)

The Mississippi River forms a 'bird's foot' type delta at the Gulf of Mexico, while Nile river forms an Arcuate delta.

4. Grand Canyon is –

- (a) a gorge
- (b) a large cannon
- (c) a river
- (d) an old cannon

43rd B.P.S.C. (Pre) 1994

Ans. (a)